Pre-class progress check

By now, you should have

1. Reviewed slides from week 2 (09.04)
2. Completed Quiz 2 on Canvas (due 8:59AM on 09.11)

1. Reviewed lab code from week 2 (09.04)

1. Checked out slides from week 3 (today)

Worked out example for bootstrap

Worked out example for permutation test

University of Pittsburgh | School of Public Health

Supervised learning (I): Regression (I)
- subset selection (best subset + stepwise)
- shrinkage (ridge + lasso + elastic net)
- dimension reduction (partial least squares)

BIOST 2155

Class 3
September 11th, 2026

Linear regression

Review

1. Simple, interpretable, and often provide an adequate description of how inputs affect an output.
2. Outperform more complex nonlinear models in situations with small training sets, low signal-to-noise ratios, or sparse data.
3. Understanding linear methods is essential for understanding complex nonlinear models.

Goal: predict a continuous output Y based on an input vector of $p-$many features.

$$\mathbf{X}^T = (X_1, X_2, \ldots, X_p)$$

$$E(Y \mid \mathbf{X}) = f(\mathbf{X}) = \beta_0 + \sum_{j=1}^{p} \beta_j X_j$$

β_j's are unknown model parameters and X_j's can be

- (Functions of) Quantitative inputs
- Numeric/dummy coding of levels of qualitative inputs
- Interactions between variables

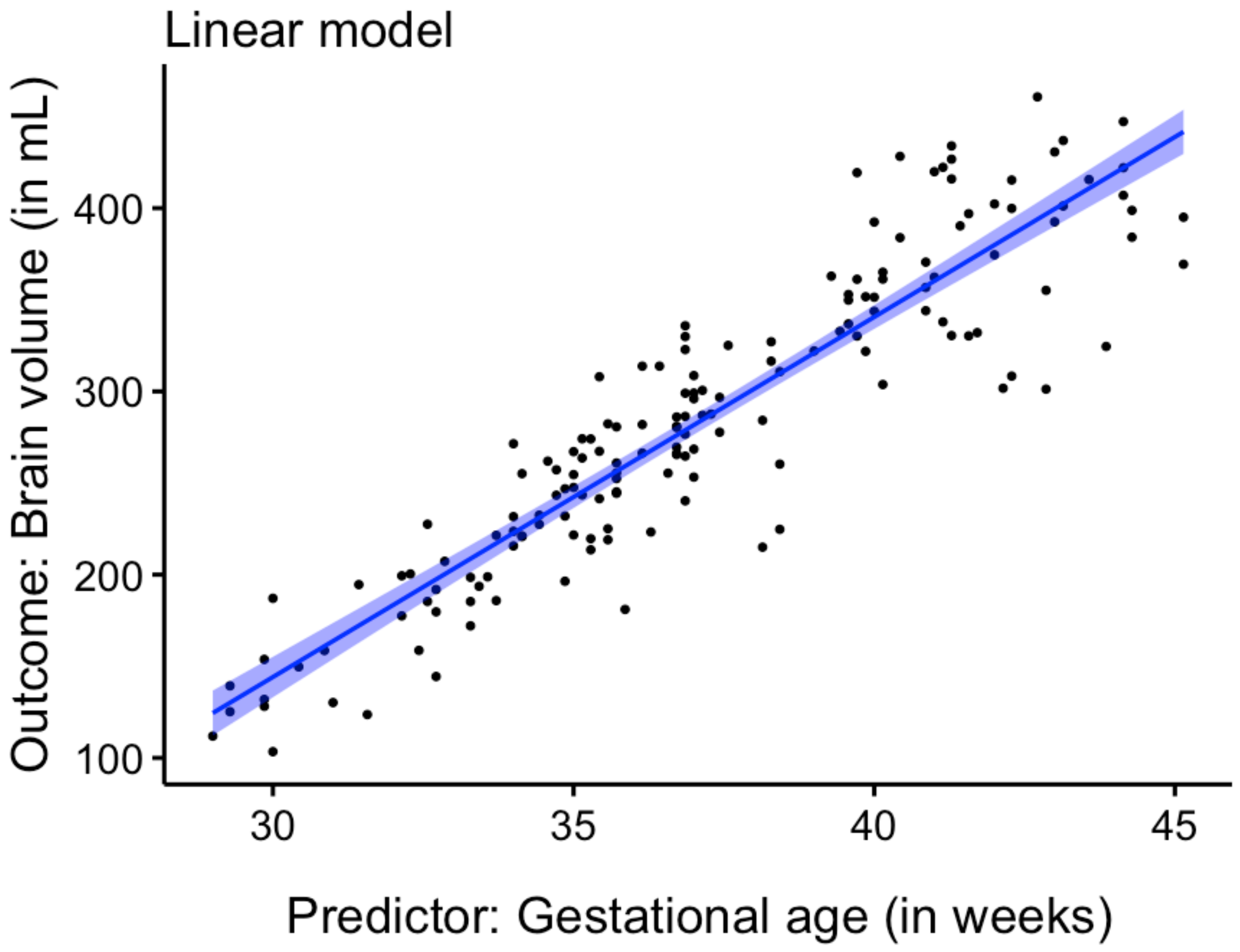
$$\mathscr{L}(\boldsymbol{X}, \boldsymbol{y}, \boldsymbol{\beta}) = RSS(\boldsymbol{\beta}) = \left(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\right)^T \left(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\right)$$

$$\hat{\boldsymbol{\beta}} = \text{argmin } RSS(\boldsymbol{\beta}) = \left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1} \left(\boldsymbol{X}^T\boldsymbol{y}\right)$$

For normally distributed error, $\hat{\boldsymbol{\beta}} \sim N\left(\boldsymbol{\beta}, \sigma^2 \left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\right)$

Standardized z-score.
Confidence interval (set) for coefficients.
F-statistic test for (group of) coefficients.

RECAP: Prediction of brain volume from gestational age



1. Fit a linear model

2. Summarize linear model
    1. Estimated model parameters
    2. Goodness of model fit

3. Testing hypothesis for model parameters

4. Confidence interval vs prediction interval

Linear regression

Review

$$RSS(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_p X_{pi}\right)^2$$

$$\hat{\boldsymbol{\beta}} = \text{argmin} \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_p X_{pi}\right)^2$$

$\hat{\boldsymbol{\beta}}$ solves the score equation: $\left.\dfrac{\partial RSS(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}\right|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}} = 0$

$$\hat{\boldsymbol{Y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\left(\boldsymbol{X}^T\boldsymbol{y}\right) = \boldsymbol{H}\boldsymbol{y}$$

$$\hat{\sigma}^2 = \sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2 / (n - p - 1)$$

Math/technical recall

Application recall

```
> summary(model)

Call:
lm(formula = `brain volume` ~ GA, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-95.486 -15.359   1.197  20.404  84.385 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -445.289     25.699  -17.33   <2e-16 ***
GA            19.647      0.688   28.56   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.63 on 160 degrees of freedom
Multiple R-squared:  0.836,	Adjusted R-squared:  0.835 
F-statistic: 815.6 on 1 and 160 DF,  p-value: < 2.2e-16

>
> confint(model, parm = "GA")
      2.5 %   97.5 %
GA 18.28867 21.00599
>
> predict(model, newdata = data.frame(1, GA = mean(data$GA)),
+         interval = "confidence")
       fit      lwr      upr
1 284.7462 279.5277 289.9647
> predict(model, newdata = data.frame(1, GA = mean(data$GA)),
+         interval = "prediction")
       fit      lwr      upr
1 284.7462 218.1206 351.3718
```

?

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???

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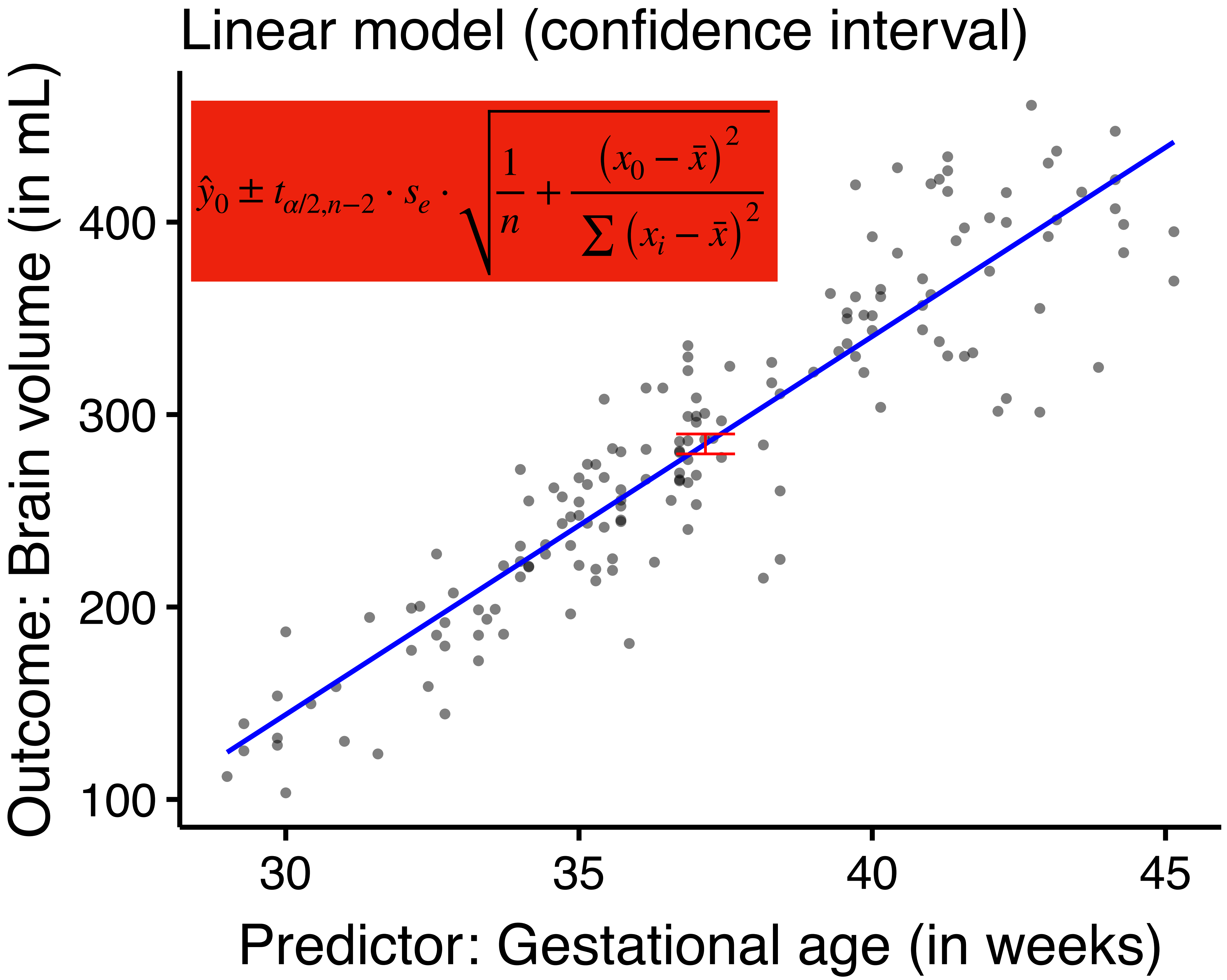
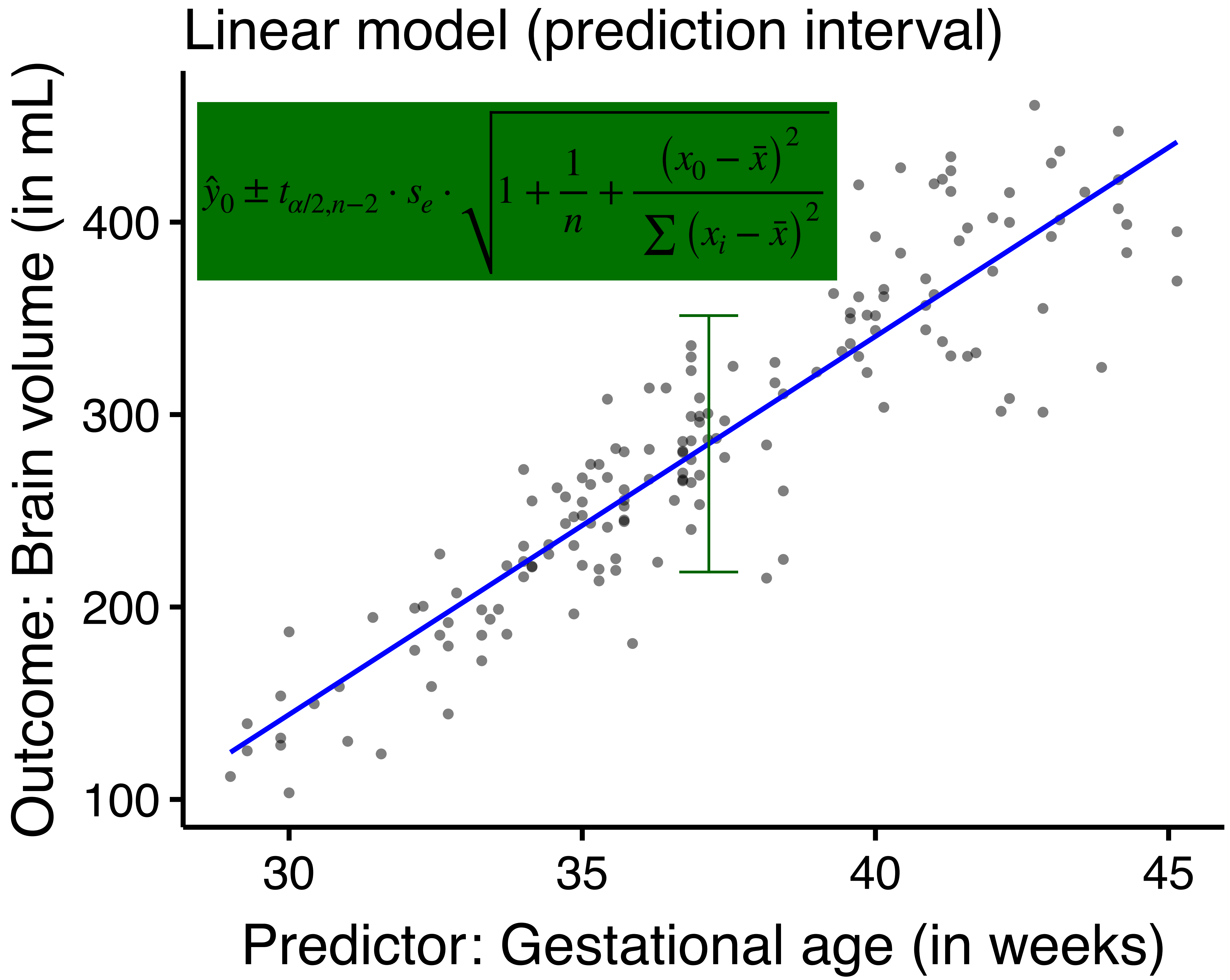
Linear regression

Review

$$s_e = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n-2}}$$





Linear models

What works and what doesn't…

- Linear models are great!
  - Interpretability
  - (Usually) good predictive performance

- Issues may arise when
  - Data are not i.i.d.
  - Collinearity
  - $p > n$

- Alternatives are a must because
  - $n \approx p$ or $p > n$ yields unstable estimates and predictions
  - Removing non-helpful covariates/predictors may improve model interpretability and generalizability.

‘Better’ linear models through…

Three classes of methods

- **Subset Selection:** identify a subset of p predictors that we believe to be related to Y and fit a model using OLS on the identified subset.
- **Shrinkage:** fit a model involving all p predictors, but $\hat{\beta}$s are shrunken towards zero relative to OLS estimates; also known as **regularization**.
- **Dimension Reduction:** project the p predictors into a M-dimensional subspace, where $M < p$, use projections as predictors to fit a linear regression model by OLS.

Stat prep for subset selection

Measures of model fit - recap

Can use k-fold CV to get error as well!
These are (computationally cheaper) other ways to estimate test error

$RSS = RSS(\hat{\beta})$ and $TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2$

Small RSS relative to TSS is a good thing!

$R^2 = 1 - \frac{RSS}{TSS}$

Large values of R^2 are good.

Adjusted $R^2 = 1 - \frac{RSS/(n-d-1)}{TSS/(n-1)}$. Large values are good.

Need to minimize $RSS/(n-d-1)$

Mallow's $C_p = \frac{1}{n}\left(RSS + 2d\hat{\sigma}^2\right)$. Small values are good.

Equivalent to AIC for normal errors.

BIC = $\frac{1}{n}\left(RSS + \log(n)d\hat{\sigma}^2\right)$. Small values are good.

BIC places heavier penalty on larger models than C_p.

*

Math prep for subset selection

Basic combinatorics

“p choose k" is a fundamental concept in combinatorics that calculates the number of ways to select a smaller group (a subset) of k items from a larger collection of p distinct items, where the order of selection doesn't matter.

1. Notation: $\binom{p}{k} = \frac{p!}{(p-k)!k!}$. Note that $k! = k \cdot (k-1)\cdots(2) \cdot 1$ and $0! = 1$.
2. How many project <u>pairs</u> can be chosen from twelve people = $\binom{12}{2}$

The total number of possible subsets of any size that can be formed from a set of p items is 2^p.

$$\sum_{k=0}^{p} \binom{p}{k} = \binom{p}{0} + \binom{p}{0} + \cdots + \binom{p}{p-1} + \binom{p}{p} = 2^p$$

Subset selection - I
Best subset selection

1. Let $\mathscr{M}_0$ denote the null model, which contains no predictors and predicts the sample mean for each observation.
2. For $k = 1,2,\ldots,p$:
   1. Fit <u>all</u> $\binom{p}{k}$ models that contain exactly k predictors.
   2. Pick the <u>best</u> (having least RSS) among these $\binom{p}{k}$ models and call it $\mathscr{M}_k$.
3. Select the best model among $\mathscr{M}_0, \mathscr{M}_1, \ldots, \mathscr{M}_{p-1}, \mathscr{M}_p$ using <u>pre-specified</u> measure of model fit.

Question 1: why only RSS? What about the other metrics?

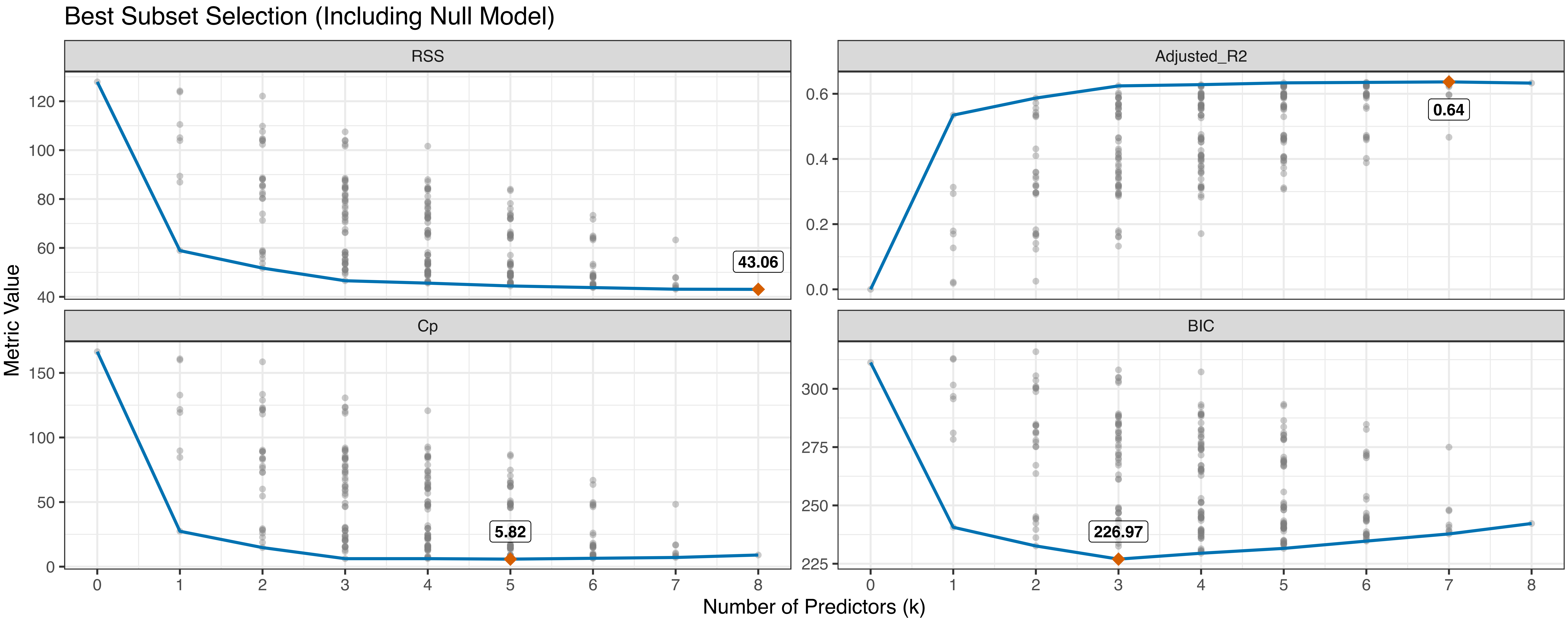
Question 2: how does the problem scale with p?

Question 3: there is a risk of overfitting for large p. Why?

Question 4: what to use instead of RSS for GLMs?

Subset selection - I

Best subset selection



Subset selection - II

Stepwise (forward) selection.

1. Let $\mathscr{M}_0$ denote the null model, which contains no predictors and predicts the sample mean for each observation.
2. For $k = 0, 1, \ldots, p-1$:
    1. Given $\mathscr{M}_k$, consider all $p-k$ (**?**) models that contain all variables in $\mathscr{M}_k$, plus one additional predictor.
    2. Pick the best (having least RSS) among these $p-k$ models and call it $\mathscr{M}_{k+1}$.
3. Select the best model among $\mathscr{M}_0, \mathscr{M}_1, \ldots, \mathscr{M}_{p-1}, \mathscr{M}_p$ using pre-specified measure of model fit.

Question 1: why only RSS?

What about the other metrics?

Question 2: how does the problem scale with p?

Question 3: not guaranteed to find the "best" model. Why?

Subset selection - III

Stepwise (backward) selection.

1. Let $\mathscr{M}_p$ denote the full model, which contains all p predictors.
2. For $k = p, (p-1), \ldots, 1$:
    1. Given $\mathscr{M}_k$, consider all k (?) models that contain all but one of the variables in $\mathscr{M}_k$.
    2. Pick the best (having least RSS) among these k models and call it $\mathscr{M}_{k-1}$.
3. Select the best model among $\mathscr{M}_0, \mathscr{M}_1, \ldots, \mathscr{M}_{p-1}, \mathscr{M}_p$ using pre-specified measure of model fit.

Question 1: why only RSS?

What about the other metrics?

Question 2: how does the problem scale with p?

Question 3: can I use backward selection when $n < p$?

Shrinkage methods

Introduction

- The subset selection methods use OLS to fit a linear model that contains a subset of the predictors.

- As an alternative, we can fit a model containing all p predictors using a technique that <u>constrains</u> or <u>regularizes</u> the coefficient estimates, or equivalently, that <u>shrinks the coefficient estimates towards zero</u>.

- Not be immediately obvious why such a constraint should improve the fit, but it turns out that <u>shrinking the coefficient estimates can significantly reduce their variance</u>.

Shrinkage methods - I

Ridge regression

The sum starts at $j = 1$.

β_0 is never penalized: shrinking the intercept toward 0 would claim the outcome is zero when every predictor is.

OLS procedure estimates $\boldsymbol{\beta} = \left(\beta_0, \beta_1, \ldots, \beta_p\right)^T$ using the values that minimizes

$$RSS(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_p X_{pi}\right)^2$$

$$\hat{\boldsymbol{\beta}}_{OLS} = \text{argmin } RSS(\boldsymbol{\beta})$$

Ridge regression estimates $\boldsymbol{\beta} = \left(\beta_0, \beta_1, \ldots, \beta_p\right)^T$ using the values that minimizes

$$RSS(\boldsymbol{\beta}) + \lambda \sum_{j=1}^{p} \beta_j^2 = \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_p X_{pi}\right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

$$\hat{\boldsymbol{\beta}}_R = \text{argmin} \left(RSS(\boldsymbol{\beta}) + \lambda \sum_{j=1}^{p} \beta_j^2 \right)$$

λ is a tuning parameter, will be determined using k-fold CV.

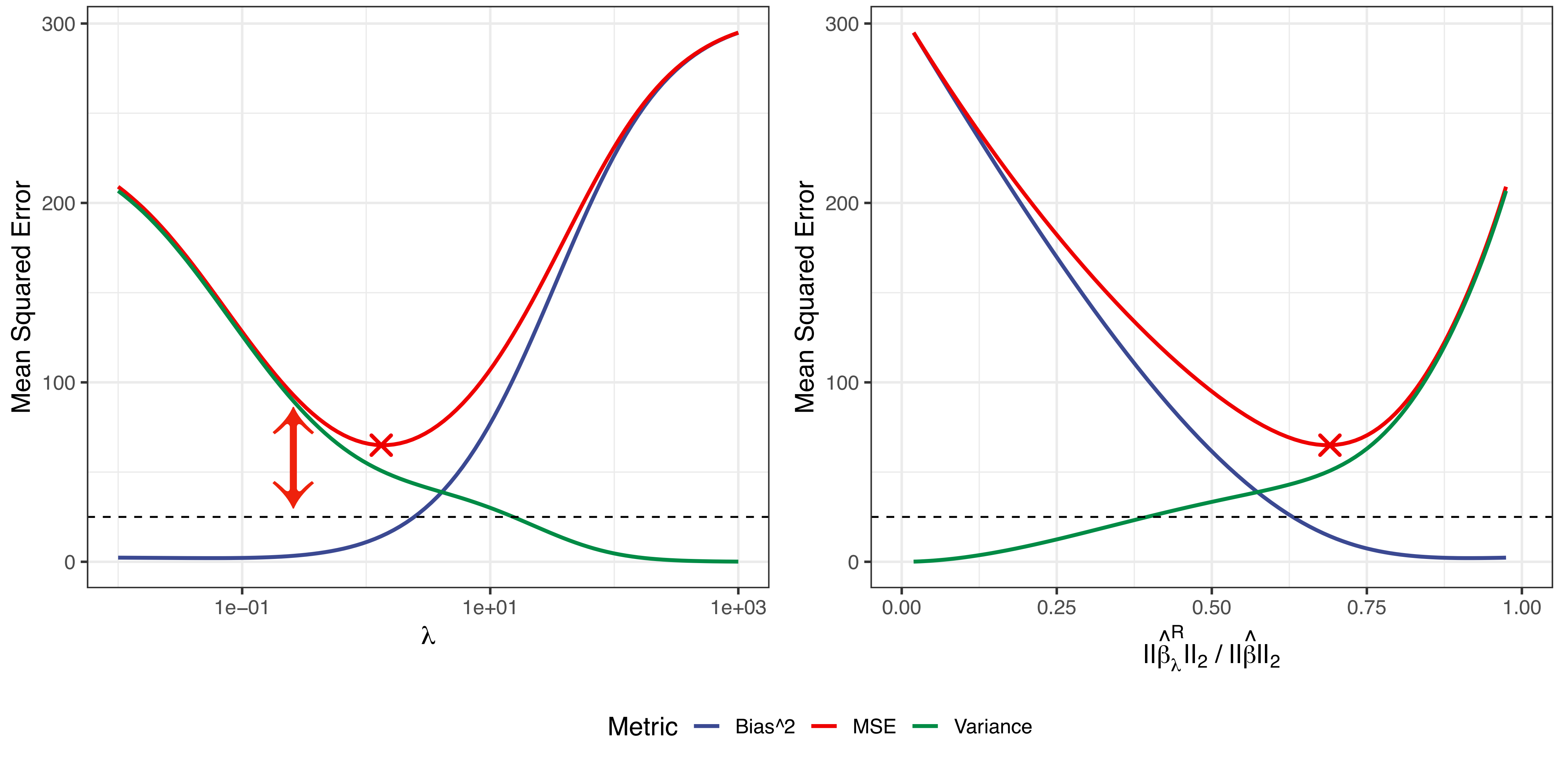
Places a **shrinkage penalty** $\lambda \sum_{j=1}^{p} \beta_j^2$ that shrinks the estimates of β_j towards zero.

Shrinkage methods - I
Ridge regression continued…

OLS estimates are scale <u>equivariant</u>: multiplying X_j by a constant c simply leads to a scaling of the $\hat{\beta}_j^{OLS}$ by $1/c$: regardless of how X_j is scaled, $X_j\hat{\beta}_j^{OLS}$ will remain the same.

In contrast, $\hat{\beta}_j^{R}$ can change substantially when multiplying a given predictor by a constant, due to the penalty part of the objective function. Therefore, it is best to apply ridge regression after **standardizing the predictors**:

$$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_{ij}-\bar{x}_j\right)^2}}.$$



Simulated data with $n = 50$ observations, $p = 45$ predictors, all having nonzero βs. Squared bias (**blue**), variance (**green**), and test mean squared error (**red**) for the ridge regression predictions on a simulated data set, as a function of λ and $||\hat{\beta}_\lambda^R||_2 / ||\hat{\beta}||_2$.

The horizontal dashed lines indicate the minimum possible MSE. The red crosses indicate the ridge regression models for which the MSE is smallest.

Increasing λ increases bias but decreases variance.

Ridge regression shrinks estimates towards zero but not exactly zero (i.e. no feature selection). When p is large, <u>all predictors contribute to the model</u> (i.e. it improves prediction accuracy but not interpretable).

Shrinkage methods - II

Lasso regression

Lasso regression estimates $\boldsymbol{\beta} = \left(\beta_0, \beta_1, \ldots, \beta_p\right)^T$ using the values that minimizes

$$RSS(\boldsymbol{\beta}) + \lambda \sum_{j=1}^{p} |\beta_j| = \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_p X_{pi}\right)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

$$\hat{\boldsymbol{\beta}}_L = \text{argmin} \left(RSS(\boldsymbol{\beta}) + \lambda \sum_{j=1}^{p} |\beta_j| \right)$$

λ is a tuning parameter, will be determined using k-fold CV.

Places a **shrinkage penalty** $\lambda \sum_{j=1}^{p} |\beta_j|$ that shrinks the estimates of β_j towards zero.

Like ridge, the lasso shrinks the $\hat{\beta}$s towards zero. However, the $\mathscr{L}_1$ penalty **forces** some $\hat{\beta}$s to be zero when λ is large.

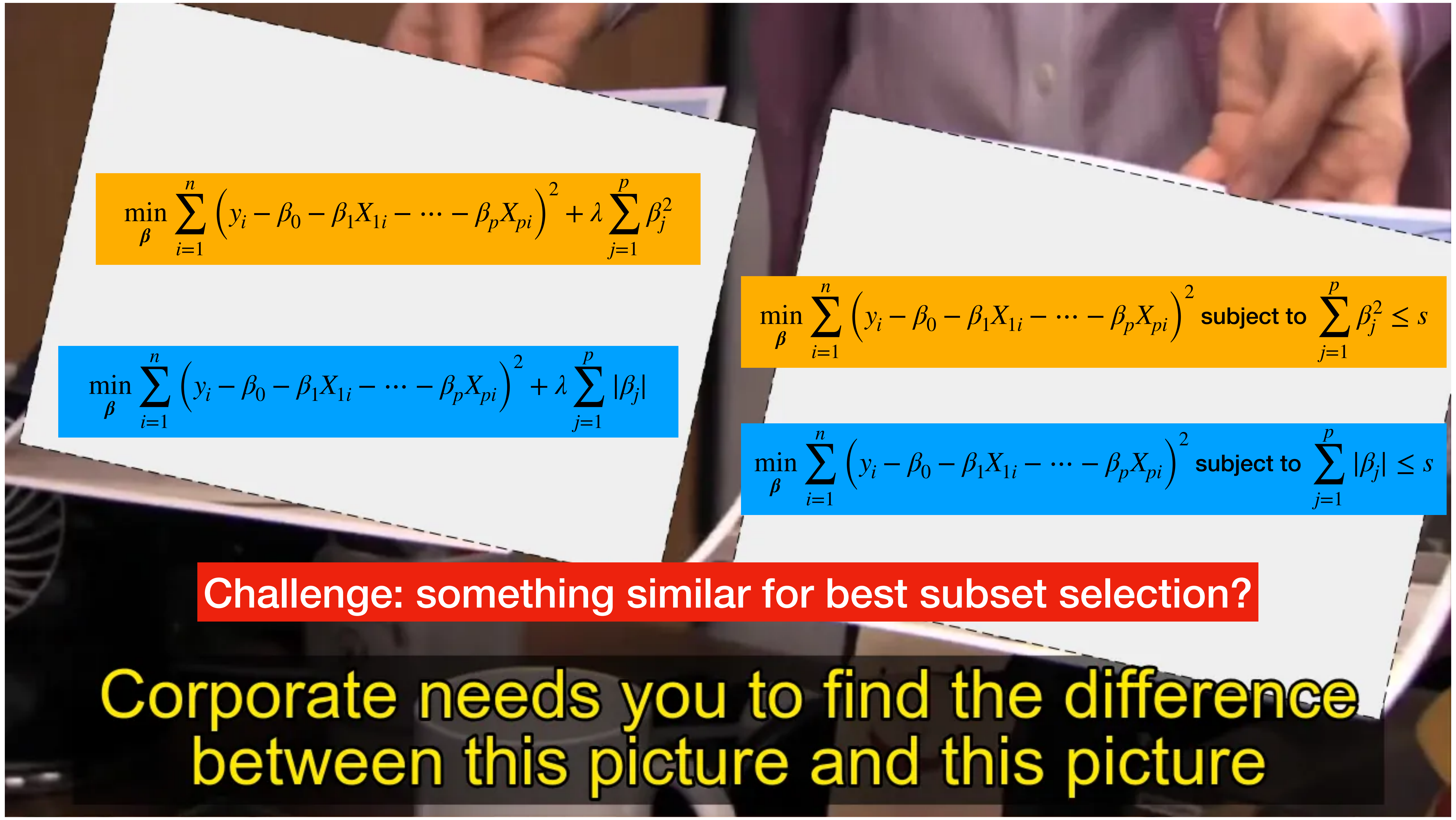
Hence, the lasso performs **variable selection** and not just **regularization.**

We say that the lasso yields **sparse** models — that is, <u>models that involve only a subset of the variables</u>.

Shrinkage methods - III

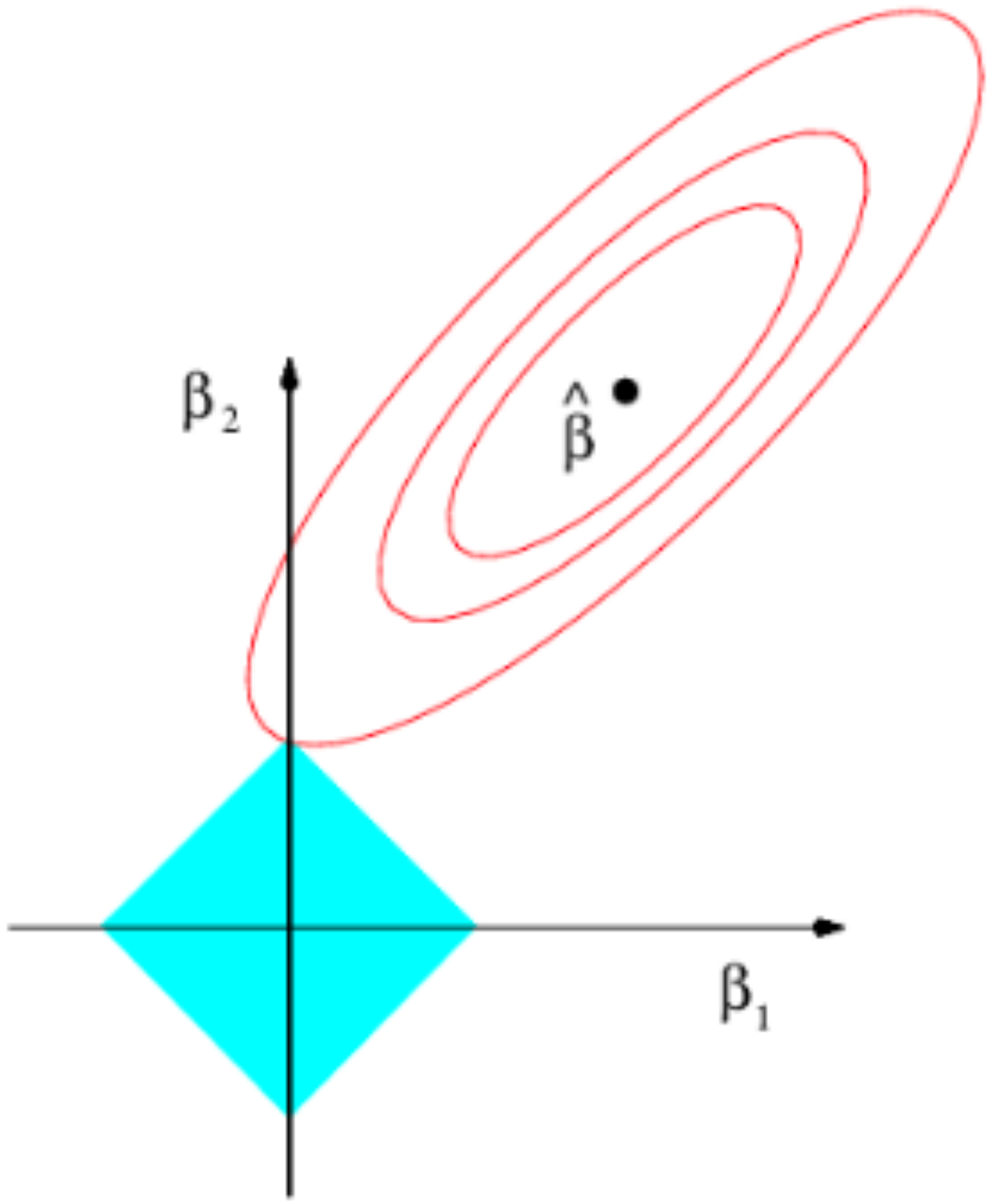
Ridge-vs-Lasso

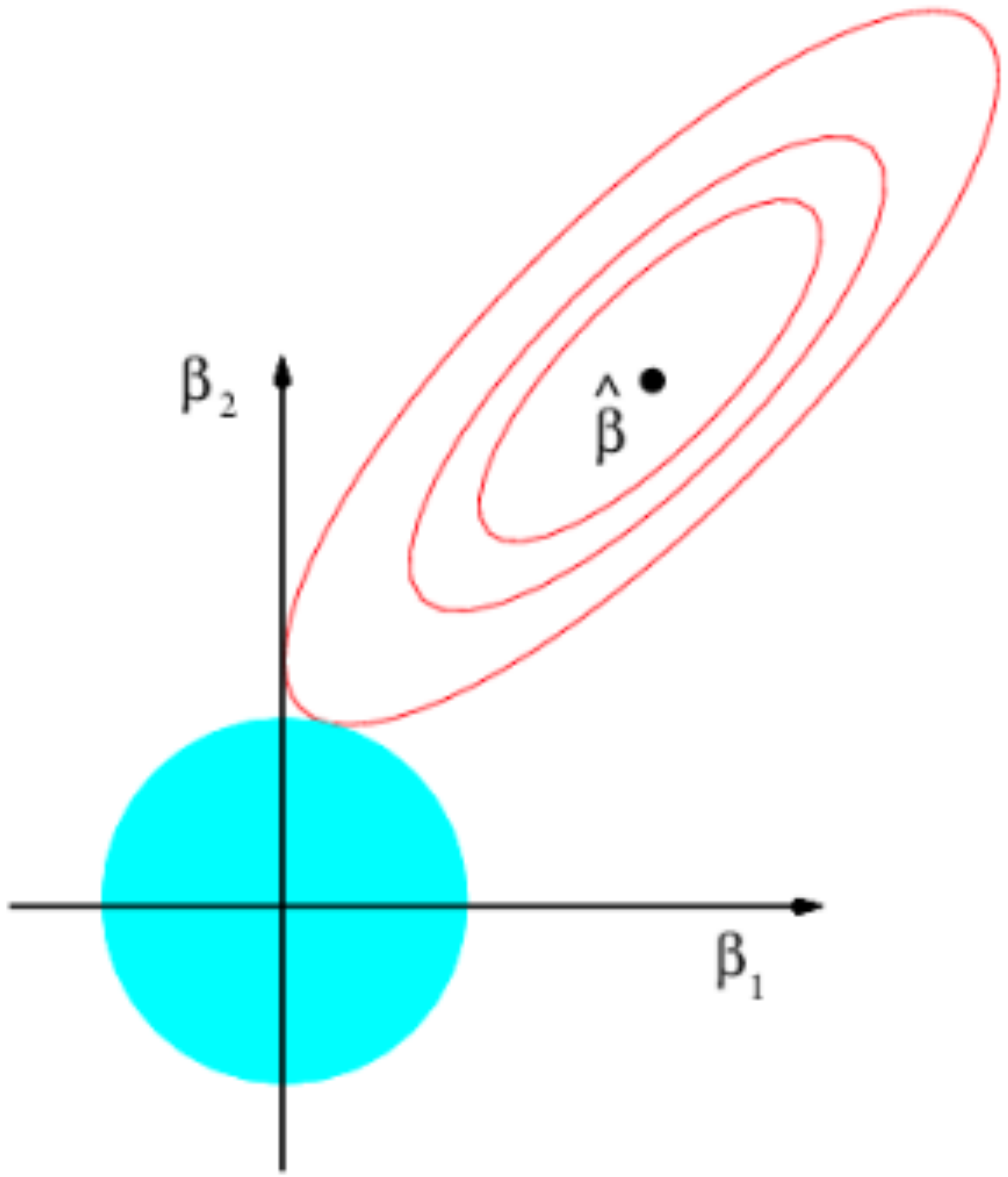
Why is it that the lasso, unlike ridge regression, results in $\hat{\beta}$s that are exactly equal to zero?



Shrinkage methods - III

Ridge-vs-Lasso



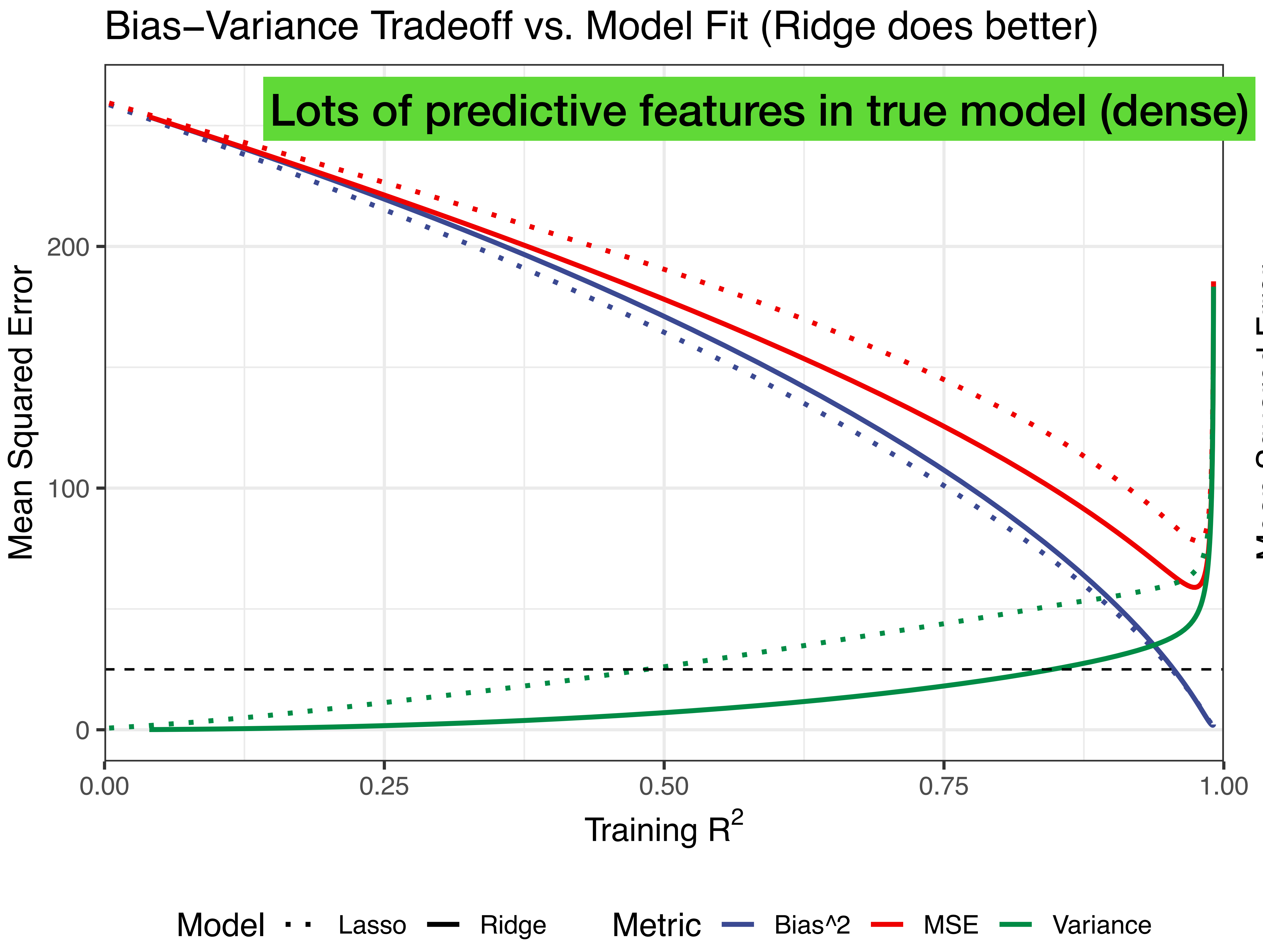


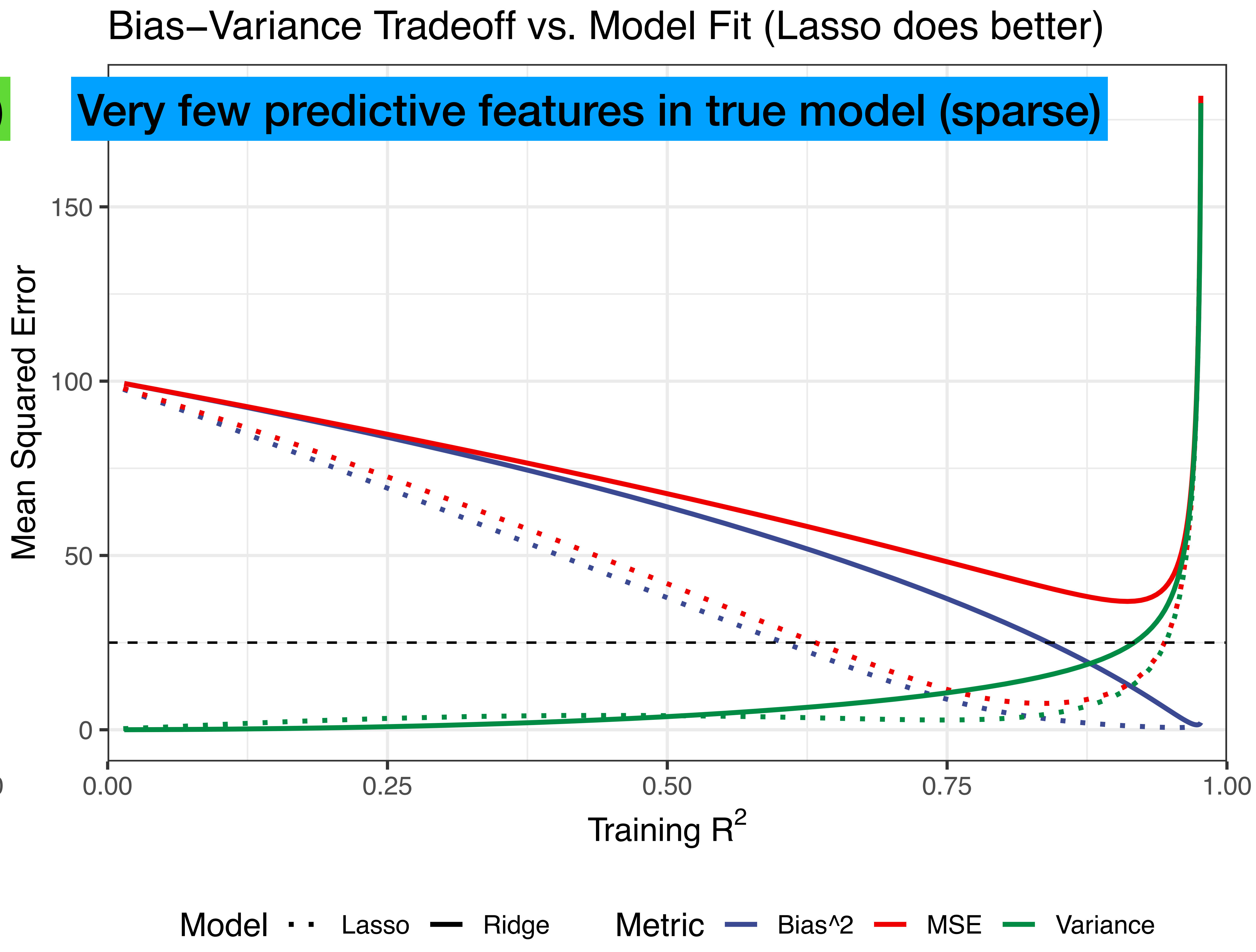
Why is it that the lasso, unlike ridge regression, results in $\hat{\beta}$s that are exactly equal to zero?

The optimizer of lasso almost always lies on a wedge, forcing some coefficient estimates to exactly 0

Shrinkage methods - III

Ridge-vs-Lasso





Demonstrating ridge and lasso: implementation and deliverables

Shrinkage methods - IV

Elastic net

- **A problem with lasso:** When two predictors are highly correlated, lasso tends to randomly pick one and ignore the other.
- **Solution:** Elastic net model, which retains a "more complete" effective feature set than lasso.

$$\hat{\boldsymbol{\beta}}_{EN} = \text{argmin}\left(RSS(\boldsymbol{\beta}) + \lambda \left[(1-\alpha)\sum_{j=1}^{p} |\beta_j| + \alpha \sum_{j=1}^{p} \beta_j^2 \right] \right)$$

- *Both α and λ are found using k-CV.* **Elastic net is computationally more costly!**

Derived input directions

Introduction (<u>will skip principal components for now</u>)

- Linear regression models using the original predictors $X_1, X_2, \ldots, X_{p-1}, X_p$ with different estimation techniques:
  - Ordinary least squares
  - Subset selection
  - Shrinkage
- Now, we'll explore approaches that transform the predictors and THEN fit a least squares model.

In many situations we have a large number of inputs, often very correlated.

In this section, we produce a small number of linear combinations $Z_m, m = 1,2,\ldots, M$ of the original inputs $X_j, j = 1,2,\ldots, p$, and Z_ms are then used in place of X_js as inputs in the regression.

$m < p$

The methods differ in how the linear combinations are constructed.

Dimension reduction (kind of…)

Math notation (🥴)

General schematic of reducing dimensionality of predictors

- **Step 1:** Let $Z_1, Z_2, \ldots, Z_M$ represent $M < p$ linear combinations (weighted average) of our original predictors:

$$\boldsymbol{Z}_m = \sum_{j=1}^{p} \phi_{mj} \boldsymbol{X}_j$$

- **Step 2:** Fit a linear regression model

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \epsilon_i \qquad \iff \qquad \boldsymbol{Y} = \boldsymbol{Z\theta} + \boldsymbol{\epsilon}$$

How well/poorly we choose $\phi_{m1}, \phi_{m2}, \ldots, \phi_{mp}$ will make/break this method.

$$\sum_{m=1}^{M} \theta_m z_{im} = \sum_{m=1}^{M} \theta_m \sum_{j=1}^{p} \phi_{mj} x_{ij} = \sum_{j=1}^{p} \sum_{m=1}^{M} \theta_m \phi_{mj} x_{ij} = \sum_{j=1}^{p} \beta_j x_{ij}$$

$$\beta_j = \sum_{m=1}^{M} \theta_m \phi_{mj}$$

<u>Potential gains</u>
Lose a little: higher bias
Overall: less MSE

Dimension reduction serves to constrain the estimated β_j coefficients

Partial least squares (PLS)

Outcome-guided specification of $\phi_{m1}, \phi_{m2}, \ldots, \phi_{mp}$

- PLS is a dimension reduction method, which
  - First identifies a new set of features $\boldsymbol{Z}_1, \boldsymbol{Z}_2, \ldots, \boldsymbol{Z}_M$ that are linear combinations of the original features
  - Then fits a linear model via <u>OLS</u> using these M new features

- PLS specifies $\phi_{m1}, \phi_{m2}, \ldots, \phi_{mp}$ by making use of BOTH $\boldsymbol{Y}$ and $\boldsymbol{X}$
  - $\boldsymbol{Z}_1, \boldsymbol{Z}_2, \ldots, \boldsymbol{Z}_M$ must be 'representative' of $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_p$
  - $\boldsymbol{Z}_1, \boldsymbol{Z}_2, \ldots, \boldsymbol{Z}_M$ must <u>also</u> be related to $\boldsymbol{Y}$

Partial least squares (PLS)

Outcome-guided specification of $\phi_{m1}, \phi_{m2}, \ldots, \phi_{mp}$

Step 1: Standardizing p predictors **and** outcome (mean = 0, variance = 1)

Step 2: To calculate $Z_1 = \sum_{j=1}^{p} \phi_{1j} X_j$, we specify $\phi_{1j} =$ lm$(Y \sim X_j)$$coefficients

Step 3 for $\boldsymbol{k = 1,2,\ldots(M-1)}$: To calculate Z_{k+1}: regress out Z_k from $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_p$ to create $\tilde{X}_1, \tilde{X}_2, \ldots, \tilde{X}_p$ - treat these as new predictors and repeat step 2 above.

Step 4: Use $\boldsymbol{Z}_1, \boldsymbol{Z}_2, \ldots, \boldsymbol{Z}_M$ as covariates to predict $\boldsymbol{Y}$

What is a ‘good’ M?

So…what do we *actually* do?

<u>Selecting</u> the right model <u>selection</u> technique

Consider an example with two correlated inputs X_1 and X_2 with correlation ρ. The true regression model is:

$Y = \beta_1 X_1 + \beta_2 X_2$ where $\beta_1 = 8$ and $\beta_2 = 2$

We will study the coefficient profiles for different methods as we vary tuning parameter, for different values of ρ.

Ridge regression shrinks all directions, but (can) shrink low-variance directions more

Partial least squares (PLS) also tends to shrink the low-variance directions, but (can) inflate some of the higher variance directions.

PLS and ridge regression (usually) tend to behave similarly.

For minimizing prediction error, ridge regression is generally preferable to variable subset selection

Ridge regression may be preferred because it shrinks smoothly, rather than in discrete jumps.

Lasso falls somewhere between ridge regression and best subset regression, and enjoys some of the properties of each.

Post-class debrief

Today we discussed…

While simple and interpretable, standard linear regression (OLS) can produce unstable estimates when predictors are numerous or correlated, potentially harming model generalizability.

Three Strategies for Better Models

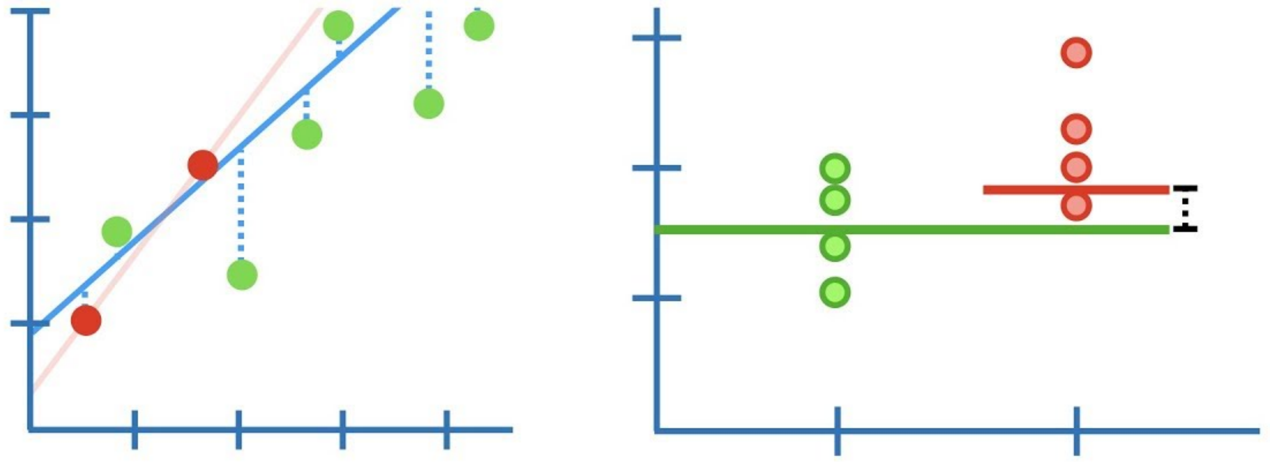
1. Subset selection
2. Shrinkage
3. Dimension reduction

1. **Subset Selection:** Identify and fit a model using only the most impactful subset of predictors. Includes best-subset, forward stepwise, and backward stepwise selection. Produces a simpler, more interpretable model by eliminating non-helpful predictors.
2. **Shrinkage (Regularization):** Fit a model with all predictors but constrain or "shrink" the coefficient estimates towards zero to reduce variance. Ridge Regression and Lasso Regression have places different penalties on coefficients.
3. **Dimension Reduction:** Project the original predictors into a smaller set of new, uncorrelated features. Partial Least Squares (PLS) creates a small number of linear combinations of the original predictors that are then used to fit an OLS model. These new features are constructed to be related to the outcome.

Before next class…

1

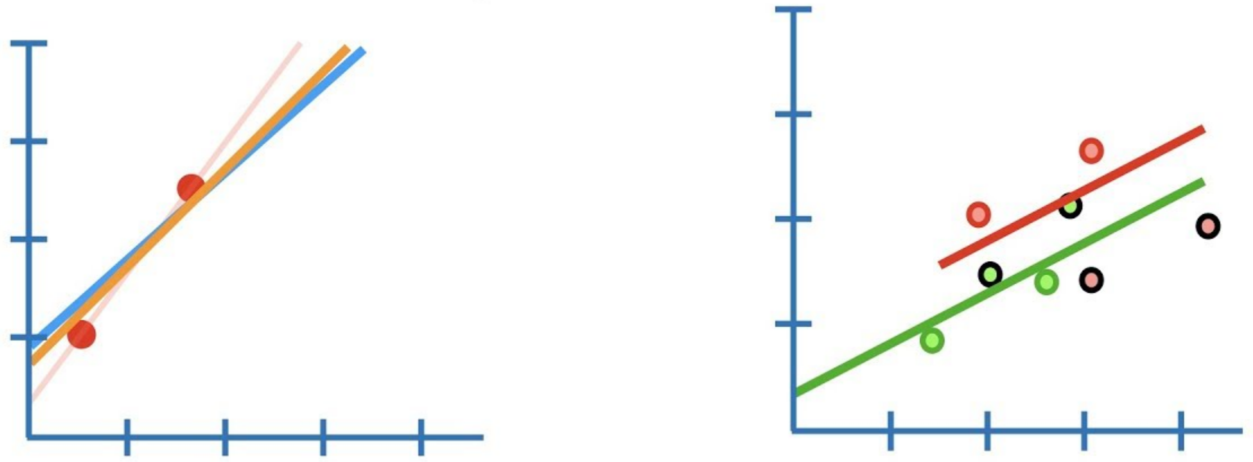
Ridge Regression….



…Clearly Explained!!!

2

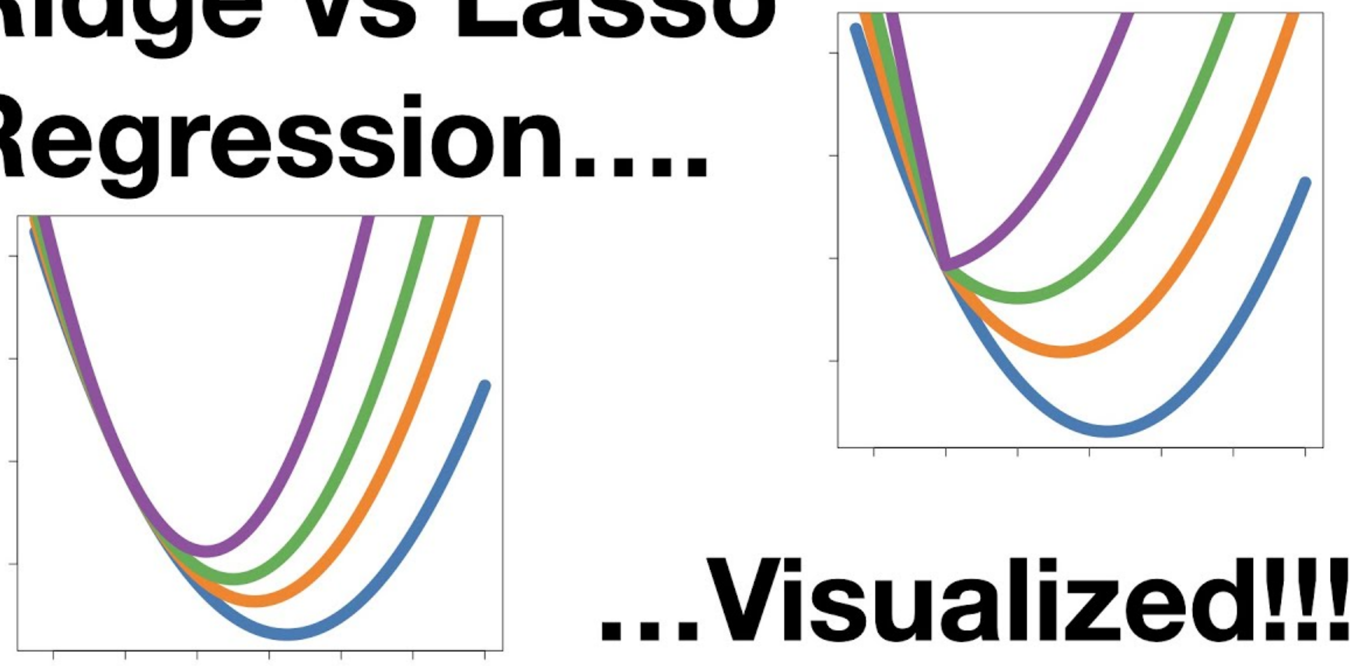
Lasso Regression….



…Clearly Explained!!!

3

Ridge vs Lasso Regression….



…Visualized!!!

4

Review ISLR 6

Review code

QUIZ 3 DUE BY 8:59AM NEXT FRIDAY SEPTEMBER 18TH

LECTURE 4 AT 9:00AM NEXT FRIDAY SEPTEMBER 18TH

HOMEWORK 1 RELEASED AT 11:00AM TODAY


That's all Folks!